7 Math Unit 0 Fractions and Decimals STAR Cards

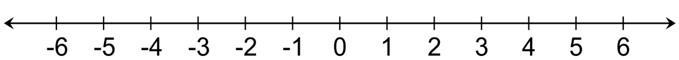
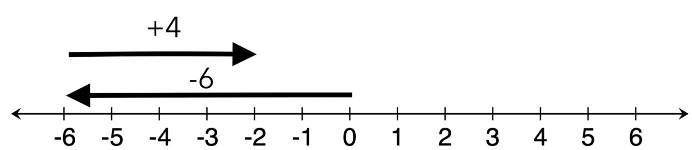
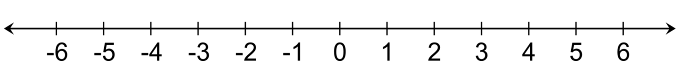
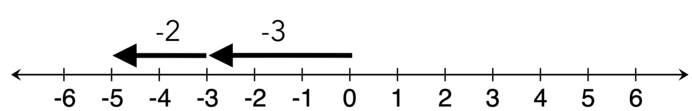
QUESTION	ANSWER
0a) ☆☆☆ What is a fraction?	0a) A fraction is a number that shows a **part of a whole**
0b) ☆☆☆ 1. What does the top number tell you? $\frac{3}{5}$ 2. What does the bottom number tell you?	0b) 1. How many **equal parts you have** (numerator) 2. How many **equal parts make up the whole** (denominator)
0c) ☆☆☆ Explain how to add (or subtract) fractions like $\frac{3}{5}+\frac{1}{5}$	0c) Because denominators are the SAME, add (or subtract) the numerators and keep the denominator. $\frac{3}{5}+\frac{1}{5} = \frac{4}{5}$

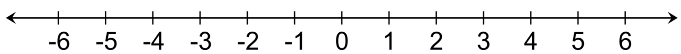
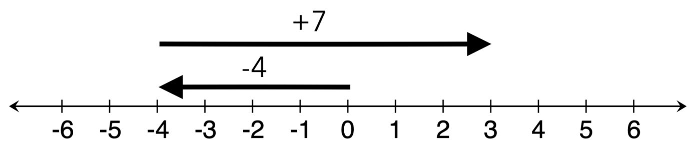
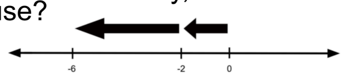
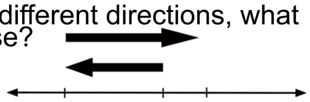
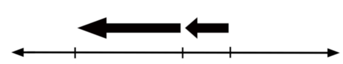
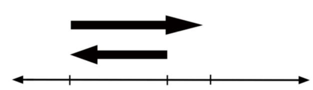
QUESTION	ANSWER
0d) Explain how to add (or subtract) fractions like $\frac{3}{5} + \frac{1}{4}$	0d) $\boxed{\frac{3}{5} + \frac{1}{4}}$ *Because denominators are DIFFERENT,* 1st **Find the LCM** 20 15 10 5 M 20 16 12 8 4 2nd **Make equal fractions** with the LCM as the denominators $\frac{3}{5} = \frac{12}{20}$ + $\frac{1}{4} = \frac{5}{20}$ 3rd **Add numerators, keep denominators** $\frac{12}{20} + \frac{5}{20} = \frac{17}{20}$
0e) Explain the rules for multiplying fractions with this example: $\frac{2}{3} \times \frac{4}{5}$	0e) **Multiply straight across** $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$ simplify, if needed
0f) 1) How do we <u>simplify fractions</u>? 2) How can you simplify $\frac{6}{15}$?	0f) 1. To simplify fractions, **divide the top and the bottom by the same number.** 2. $\frac{6}{15} \frac{\div 3}{\div 3} = \frac{2}{5}$

QUESTION	ANSWER
0g) How do we convert a mixed number into an improper fraction? **Example:** $4\frac{1}{3}$	0g) 2nd) Add the numerator $\overset{+1}{} \quad \overset{\times 3}{}$ $4\frac{1}{3} = \frac{13}{3}$ 1st) Multiply the whole number by the denominator (=12) whole number by denominator = 12 3rd) Keep the same denominator
0h) How do we change an improper fraction into a mixed number? **Example:** $\frac{10}{7}$	0h) Divide the numerator (top) by the denominator (bottom). Write your remainder as a fraction. $\begin{array}{r} 1 \\ 7\overline{)10} \\ -7 \\ \hline 3 \end{array} = 1\frac{3}{7}$
0i) How do we <u>divide fractions</u>? Example: $\frac{1}{4} \div \frac{3}{8}$	0i) To divide fractions $\frac{1}{4} \div \frac{3}{8}$ **K**eep **C**hange **F**lip ↓ ↓ ↓ $\frac{1}{4} \times \frac{8}{3}$ *then multiply straight across* $\frac{1}{4} \times \frac{8}{3} = \frac{8}{12}$ *or* $\frac{2}{3}$

7 Math Unit 1 Integer Operations STAR Cards

1

QUESTION	ANSWER
1a) ☆☆☆ Before solving, how would you simplify these problems? 1. $5 + \ -3$ 2. $-7 - \ -2$	1a) 1. $5 + \ -3$ simplifies to $5 - 3$ 2. $-7 - \ -2$ simplifies to $-7 + 2$
1b) ☆☆☆ **Use a number line** to model this problem: $-6 + 4$ 	1b) $-6 + 4$  *The answer is -2*
1c) ☆☆☆ Use a number line to model this problem: $-3 + \ -2$ 	1c) $-3 + \ -2$  *The answer is -5*

QUESTION	ANSWER
1d) Use a number line to model this problem: $-4 - \quad -7$ 	1d) $-4 - \quad -7$ \\  *The answer is 3*
1e) Following the rules for multiplying and dividing with negatives, are these answers **positive** or **negative**? 1. $-\frac{1}{2}$ x $\frac{3}{4} =$ 2. $-a$ x $-b =$ 3. $4.8 \div -1.2 =$ 4. $323 \div 2 =$	1e) **Multiply and divide normally.** If signs match, positive. If signs don't match, negative. 1. $-\frac{1}{2}$ x $\frac{3}{4} =$ **NEGATIVE** 2. $-a$ x $-b =$ **POSITIVE** 3. $4.8 \div -1.2 =$ **NEGATIVE** 4. $323 \div 2 =$ **POSITIVE**
1f) When arrows point the same way, what operation do you use?  When arrows point different directions, what operation do you use? 	1f) Same Direction: ADD lengths  Different Directions: SUBTRACT lengths 

7 Math Unit 2 Expressions STAR Cards

QUESTION	ANSWER
2a) 1. What do the absolute value lines $\lvert\ \rvert$ mean? 2. How are the following problems different? $\lvert -2+4 \rvert$ vs $\lvert -2 \rvert + 4$	2a) 1. Distance from zero, ALWAYS POSITIVE 2. For $\lvert -2+4 \rvert$ **:Solve the inside**, then take the absolute value. For $\lvert -2 \rvert + 4$ **:Find the absolute value of -2**, then solve.
2b) How can you simplify an expression like this one: $4x + 3 + 2x + 5$	2b) **Combine like terms** Same variables go together, Normal numbers *go* together $4x + 3 + 2x + 5$ ↓ $6x + 8$

7 Math Unit 3 Equations STAR Cards

<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr>
<td>3a)

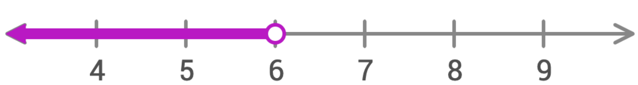
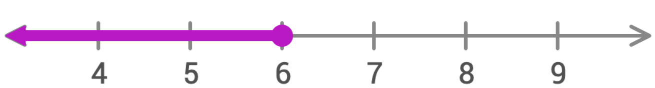
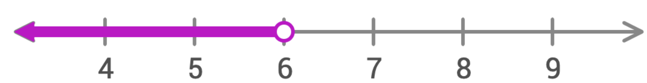
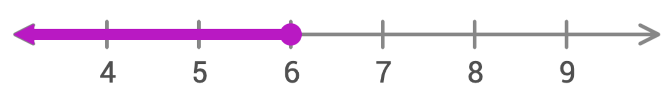
What does it mean to <u>solve</u> an equation?</td>
<td>3a)

Solving an equation means finding the value of the variable that will make the left side equal the right side.</td>
</tr>
<tr>
<td>3b)

List the inverse operations for each:
<table>
<tr><th>I see</th><th>To undo it</th><th>I do</th></tr>
<tr><td>Addition</td><td></td><td></td></tr>
<tr><td>Subtraction</td><td></td><td></td></tr>
<tr><td>Multiplication</td><td></td><td></td></tr>
<tr><td>Division</td><td></td><td></td></tr>
</table>
</td>
<td>3b)
<table>
<tr><th>I see</th><th>To undo it</th><th>I do</th></tr>
<tr><td>Addition</td><td colspan="2">Subtraction</td></tr>
<tr><td>Subtraction</td><td colspan="2">Addition</td></tr>
<tr><td>Multiplication</td><td colspan="2">Division</td></tr>
<tr><td>Division</td><td colspan="2">Multiplication</td></tr>
</table>
</td>
</tr>
<tr>
<td>3c)

How do you solve a 2 step equation like this?
$6x + 3 = 27$</td>
<td>3c)
1st) Subtract 3 from both sides
2nd) Divide both sides by 6

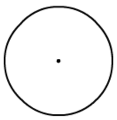
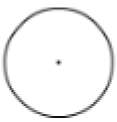
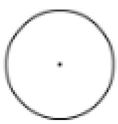
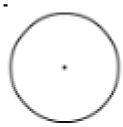
$$\begin{array}{c|c|c} 6x + 3 & = & 27 \\ -3 & & -3 \\ \hline \frac{6x}{6} & & \frac{24}{6} \\ x & = & 4 \end{array}$$</td>
</tr>
</table>

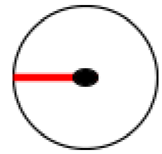
QUESTION	ANSWER
3d) How do you read each of these inequality symbols? 1. < 2. > 3. ≤ 4. ≥	3d) 1. Less than < 2. Greater than > 3. Less than or equal to ≤ 4. Greater than or equal to ≥
3e) What is the difference between an inequality and an equation? $4x + 7 \geq 20$	3e) Inequalities have lots of answers; equations only have one answer.
3f) Write an inequality for each number line below: 1.  2.  3. 	3f) 1. $x < 6$  2. $x \leq 6$  3. $x > 8$ 

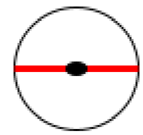
7 Math Unit 4 Circles STAR Cards

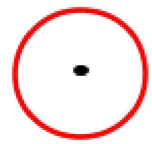
<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr>
<td>
4A)

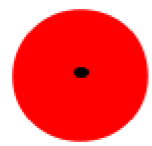
<u>Define</u> or <u>draw</u> the following parts of a circle:

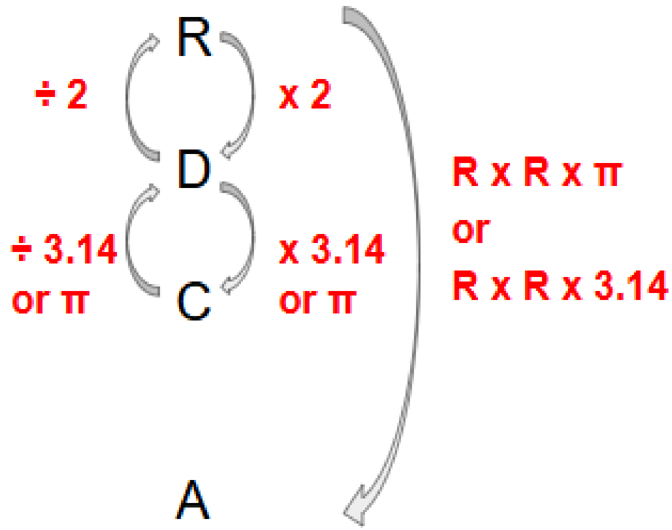
1. Radius:

2. Diameter:

3. Circumference: 
4. Area:

</td>
<td>
4A)

1. <u>Radius</u>:
center to the outside 

2. <u>Diameter</u>: Side to side through the center. 

3. <u>Circumference</u>: the outside “perimeter” of a circle. 

4. <u>Area</u>: the inside of a circle. 
</td>
</tr>
</table>

QUESTION	ANSWER
4b) ☆☆☆ Fill in the circle tool: R D C A	4b) 
4c) ☆☆☆ What are two ways you can write your answer with **pi**?	4c) 1. By using the symbol π *ex)* 10π 2. Estimated with **3.14** *ex)* $10 \times 3.14 = 31.4$

QUESTION	ANSWER
4d) Explain how you round a number to the hundredths place **8.562**	4d) 1st: Find the hundredths place *2nd:* Look one digit to the right *3rd:* If that digits **5 or more up the score,** 4 or less let it rest ↓ 8.5<u>6</u>2 Rounded = 8.56

7 Math Unit 5 Proportional Relationships STAR Cards

<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>5a) ☆☆☆

What is a proportional relationship?</td><td>5a)

A relationship where you are constantly multiplying by the same number.</td></tr>
<tr><td>5b)

The table below is proportional. How do you find the Constant Of Proportionality in a table?

How do you find the missing y value in this table?

<table>
<tr><th>Hours worked (x)</th><th>Dollars Earned (y)</th></tr>
<tr><td>2</td><td>10</td></tr>
<tr><td>6</td><td>y</td></tr>
<tr><td>x</td><td>80</td></tr>
</table>

How do you find the missing x value in this table?</td><td>5b)

1. To find the C.O.P, divide $\frac{y}{x}$

2. 6 x C.O.P = y

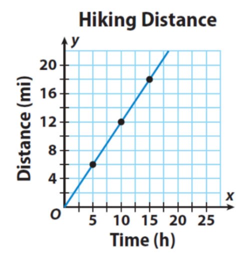
3. 80 ÷ C.O.P = x</td></tr>
</table>

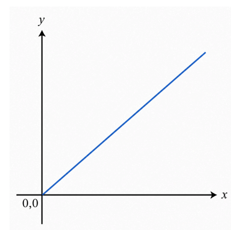
<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>5c) ☆☆☆

1. Sketch or describe what a proportional graph looks like

2. Describe how to find the C.O.P from a proportional graph</td><td>5c)

1. A straight line that goes through the origin (0,0)





2. Pick a point on the line and divide y by x (hint: when x=1, this is easiest!)</td></tr>
<tr><td>5d) ☆☆☆

Where can you find the Constant of Proportionality in an equation?

$y = \frac{1}{5}x$</td><td>5d)

The C.O.P is the number in front of the variable

$y = \frac{1}{5}x$</td></tr>
<tr><td>5e) ☆☆☆

What are 3 main ways to represent a proportional relationship?</td><td>5e)

<u>G</u>raph, <u>E</u>quation, <u>T</u>able</td></tr>
</table>

6

7 Math Unit 6 Percents STAR Cards

<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>6a)

1. What is the definition of percent?

2. How do you write 34% as a fraction?</td><td>6A)

1. A percent is a fraction out of 100

2. 34% is $\frac{34}{100}$</td></tr>
<tr><td>6b)

How can you find the percent of a number?

Ex: Find 20% of $80</td><td>6b)

Change the percent into a decimal and multiply it by the number.

Ex:
<table><tr><td>PART</td><td>20</td><td>.20</td><td>16</td></tr><tr><td>WHOLE</td><td>100</td><td>1</td><td>80</td></tr></table></td></tr>
</table>

<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr>
<td>6c)

If you need to find a percent in a real life situation how do you do it?

<u>Example:</u> Carlos got 17 out of 20 questions right on his science test. What percent did he get right?</td>
<td>6c)

1st) Find the fraction

2nd) Part divided by whole

3rd) Decimal times 100

<table>
<tr><td>PART</td><td>17</td><td>.85</td><td>85%</td></tr>
<tr><td>WHOLE</td><td>20</td><td>1</td><td>100</td></tr>
</table>
</td>
</tr>
<tr>
<td>6d)

Below is a problem where you need to find a percent increase. Explain how to solve it:

You go to a restaurant and your food costs $30. You want to add a 15% tip. What is the total amount you will pay?</td>
<td>6d)
<u>To calculate % increase:</u>

1st: Find the percent of the number

<table>
<tr><td>PART</td><td>15</td><td>.15</td><td>4.50</td></tr>
<tr><td>WHOLE</td><td>100</td><td>1</td><td>30</td></tr>
</table>
↓

2nd Add whole + % of the number

$30 + $4.50 = $34.50</td>
</tr>
</table>

<table>
<tr><th>QUESTION</th><th>ANSWER</th></tr>
<tr>
<td>6e)

How do you find the total after a % decrease?

Example: 20% discount on $30</td>
<td>6e)

<u>To calculate % decrease:</u>
1st) Find the percent of the number
↓
<table>
<tr><td>PART</td><td>20</td><td>.20</td><td>6</td></tr>
<tr><td>WHOLE</td><td>100</td><td>1</td><td>30</td></tr>
</table>
2nd) Subtract whole – $\boxed{\text{\% of the number}}$

$30 - $6 = $24</td>
</tr>
<tr>
<td>6f)

How do you convert a ….

Fraction to Decimal:

Decimal to Percent:

Percent to Fraction:</td>
<td>6f)

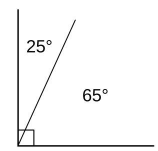
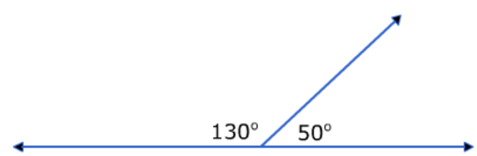
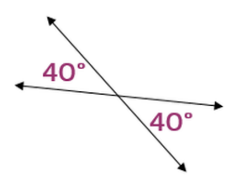
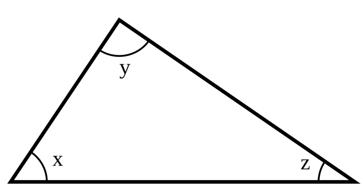
How do you convert a ….

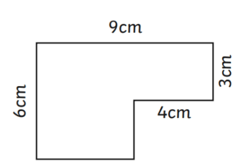
Fraction to Decimal: Numerator ÷ Demoniator

Decimal to Percent: Decimal ✖ 100

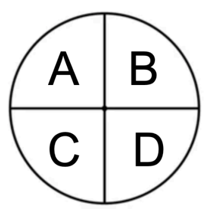
Percent to Fraction: Percent over 100</td>
</tr>
</table>

7 Math Unit 7 Geometry STAR Cards

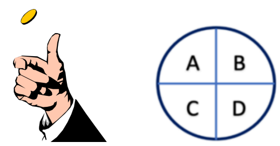
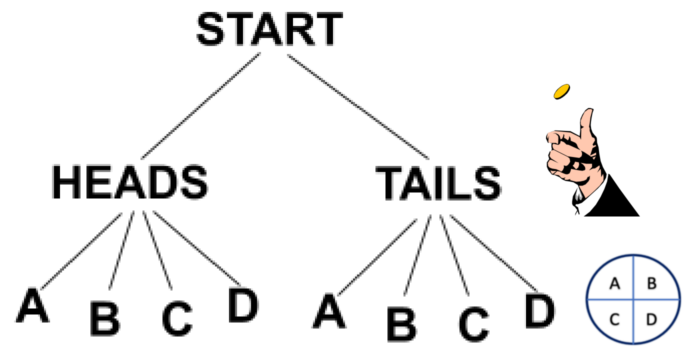
QUESTION	ANSWER
7a) 1.) Describe and sketch a complementary angle 2.) Describe and sketch a supplementary angle. 3.) Describe and sketch a vertical angle.	7a) 1. Complementary angles **add up to 90°**  2. Supplementary angles **add up to 180°**  3. Vertical angles are **opposite angles that are equal**. 
7b) What do the interior angles of any triangle in the world add up to? 	7b) The three angles in any triangle add up to 180. $x + y + z = 180°$

QUESTION	ANSWER
7c) How can you tell if three **lines** can form a triangle? Like these lines:	7c) The two smallest sides added together have to be bigger than the longest side. These lines would form a triangle!
7d) What is the difference between volume and surface area?	7d) Volume = The number of cubes that fit **inside** a **3D shape** Surface Area = The area of all the flat sides of **a 3D shape**
7e) How do you calculate the area of… • a rectangle?  • a triangle?  • a compound shape? 	7e) To find the area of a… ● Rectangle: **Length x Width** ● Triangle: **Base x Height ÷ 2** ● Compound shapes: 1.Break into shapes you know 2.Find the area of each 3.Add all areas together

7 Math Unit 8 Probability and Statistics STAR Cards

QUESTION	ANSWER
8a) 1. What is probability? 2. How do you find probability?	8a) 1. Probability is the **chance** something will happen 2. To find probability, write a fraction: $\frac{\text{Desired outcomes}}{\text{Total outcomes}}$
8b) If you spin this spinner 40 times how can you **predict how many times (probability)** you'll land on C?  	8b) Multiply the **Probability x total tries** $\frac{1}{4}$ x 40= 10 times
8c) 1. What is the difference between a <u>population</u> and a <u>sample?</u> 2. Why take a sample instead of surveying the whole population?	8c) 1. <u>POPULATION</u>: The **whole group** you are studying <u>SAMPLE</u>: a **small group** selected from the population 2. Surveying a sample is more efficient!

QUESTION	ANSWER
8d) What makes a <u>sample</u> **representative** and not **biased**?	8d) Representative samples: 1. Sample is **FROM the population** 2. Sample is as **BIG** as possible 3. Sample is **RANDOM** *Biased samples are not fair or favor one group*
8e) 1) What is a <u>measure of center</u>? Give 2 examples 2) What is a <u>measure of variability</u>? Give 2 examples	8e) 1) <u>Measures of center:</u> represent the **typical value** of the data set *Examples:* **mean, median, mode** 2) <u>Measure of variability:</u> measures how spread out your data is. *Examples:* **Range, MAD, IQR**
8f) 1. How do you calculate Mean/Average? 2. How do you calculate Median?	8f) 1) To calculate Mean(average): **1st) Add all the numbers** **2nd) Divide by how many numbers you have** 2) To calculate median: 1st) Put the numbers in **order** from **least to greatest** 2nd) Choose the **middle** number *If there is a tie between two numbers in the middle, add them and divide by 2*

QUESTION	ANSWER
8g) You win a game if you: ● Flip a coin to heads ● Spin this spinner and land on A  Sketch a tree diagram and write how many possible outcomes there are in this situation.	8g)  There are 8 possible outcomes